

COURSE SYLLABUS

AI Essentials

COURSE NUMBER

AIE-001

DURATION

1 day · 4 modules

AUDIENCE LEVEL

Beginner

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This course teaches the fundamental concepts of artificial intelligence, machine learning, and generative AI technologies. It is designed for business professionals and non-technical managers with no prior programming or computer science experience. Afterward, learners will be able to identify AI opportunities within their organizations, write effective prompts, and ethically implement AI tools to improve daily productivity.

DELIVERY FORMAT OPTIONS

Standard Instructor-Led Training (ILT). The instructor teaches each module, leads explanations and discussions, and guides students through the standard module labs using the published module timing.

Bootcamp / Lab-Centered Delivery. When Bootcamp Mode capstone labs are included, they provide an alternative delivery configuration, not additional required ILT lecture time. In this configuration, students primarily work through module labs and capstone lab scenarios while the trainer acts as a facilitator, answering questions, clarifying concepts, and helping students work through lab challenges.

Training providers may choose either the normal ILT model or the Bootcamp/lab-centered model based on audience readiness, delivery goals, and available class time.

AT COURSE COMPLETION

After completing this course, you will be able to:

Distinguish between artificial intelligence, machine learning, and deep learning paradigms.

Formulate effective prompts using structured engineering techniques to generate high-quality text and image outputs.

Evaluate business use cases to identify high-value opportunities for AI integration.

Analyze the ethical implications, bias risks, and data privacy concerns associated with enterprise AI adoption.

Formulate a personal productivity plan leveraging accessible web-based AI assistants.

PREREQUISITES

- Basic computer literacy and familiarity with standard web browsers and office productivity software.
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COURSE OUTLINE

MODULE 1 Demystifying Artificial Intelligence and Machine Learning Foundations

85 min

Understand the defining characteristics of artificial intelligence, machine learning, and deep learning. Explore how modern foundation models operate and what distinguishes generative technologies from classic algorithmic automation.

LESSONS

LAB EXERCISES

Lesson 1: The Spectrum of Intelligence: AI, ML, and DL	— 15 min	Analyzing AI Outputs and Model Capabilities	— 30 min
Lesson 2: Understanding Generative AI and Large Language Models	— 15 min		
Lesson 3: Key Terminology: Neural Networks, Foundation Models, Parameters, And Tokens	— 15 min		

TIMING

Review / Summary: 10 min · 4 questions **Module total:** 85 min

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Distinguish between artificial intelligence, machine learning, and deep learning paradigms.
- Explain how generative AI and large language models process data and generate original responses.
- Contrast classic deterministic programming rules with modern probabilistic, neural network-based pattern recognition.

MODULE 2 Practical Prompt Engineering for Generative AI Tools 100 min

Learn to write clear, structured, and deliberate instructions that shape the context, tone, and format of AI outputs. Master repeatable frameworks to streamline daily messaging and document-generation tasks.

LESSONS

Lesson 1: Elements of a Structured Prompt: Context, Persona, and Constraints — 15 min

Lesson 2: Advanced Prompting Patterns: Few-Shot, Chain-of-Thought, and Role-Playing — 15 min

Lesson 3: Techniques For Iterative Prompting And Maintaining Conversational State — 25 min

LAB EXERCISES

Engineering Multi-Turn Prompts for Workplace Workflows — 35 min

TIMING

Review / Summary: 10 min · 4 questions **Module total:** 100 min

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Formulate effective prompts using structured engineering techniques to generate high-quality text and image outputs.

Implement advanced prompting structures like role-play parameters, system contexts, and formatting rules.

Analyze output failures to revise and iteratively refine multi-turn prompts for accuracy.

MODULE 3 Identifying and Evaluating Enterprise AI Use Cases

85 min

Discover how to identify workflows inside your business units that are prime targets for AI integration. Master structural frameworks to evaluate and score these potential initiatives.

LESSONS

Lesson 1: Mapping AI Capabilities to Business Challenges — 15 min

Lesson 2: Feasibility and Impact Assessment Matrix — 15 min

Lesson 3: The Feasibility Vs. Business Value Priority Matrix — 15 min

LAB EXERCISES

Evaluating and Prioritizing Organizational AI Opportunities — 30 min

TIMING

Review / Summary: 10 min · 4 questions **Module total:** 85 min

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Evaluate business use cases to identify high-value opportunities for AI integration.

Map standard operational business workflows to appropriate categories of generative and analytical AI capabilities.

Assess potential workflows using objective criteria to determine implementation feasibility and ROI.

MODULE 4 Navigating AI Ethics, Data Privacy, and Security in the Workplace

90 min

Examine critical security, compliance, and ethical standards that must govern your professional AI usage. Learn how to securely configure tools to protect sensitive company data.

LESSONS

Lesson 1: Algorithmic Bias, — 15 min
Hallucination, and
Intellectual
Property Risks

Lesson 2: Data Privacy, — 15 min
Corporate
Compliance, and
Secure Tool
Configuration

Lesson 3: Data Leakage — 20 min
Risks: How Public
LLM Systems
Consume User
Training Inputs

LAB EXERCISES

Drafting an Acceptable Use — 30 min
and Data Security Policy
Checklist

TIMING

Review / Summary: 10 min · 4 questions **Module total:** 90 min

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Analyze the ethical implications, bias risks, and data privacy concerns associated with enterprise AI adoption.

Formulate a personal productivity plan leveraging accessible web-based AI assistants safely.

Apply data classification and privacy configurations to block corporate data leaks while using public LLMs.

BOOTCAMP MODE — CAPSTONE LABS

Capstone labs are included for Bootcamp/lab-centered delivery. They are not additional required lecture time for Standard Instructor-Led Training. Use them when the client chooses a facilitated lab delivery model where the trainer coaches students through integrated practice scenarios.

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3, 4

SKILLS TO BE INTEGRATED

Interactive LLM system performance evaluation and comparative capability profiling.

Developing context-aware, structured role-prompts leveraging clear target instructions and formatting constraints.

Scoring enterprise business problems using feasibility matrices and impact prioritization matrices.

Auditing draft guidelines to balance employee productivity against corporate compliance risks.

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Courseware Publisher download package.